

Lab Sheet 3

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Section: B

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```
In [1]: import cv2
import numpy as np
import matplotlib.pyplot as plt
import urllib.request
import os

print("Libraries imported successfully!")
```

Libraries imported successfully!

```
In [3]: image_url = "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg"
image_path = "lena.jpg"

urllib.request.urlretrieve(image_url, image_path)

print("Dataset downloaded successfully!")
```

Dataset downloaded successfully!

```
In [4]: image = cv2.imread(image_path)

if image is None:
    print("Error: Image not loaded")
else:
    print("Image loaded successfully!")
    print("Image Shape:", image.shape)
```

Image loaded successfully!
Image Shape: (512, 512, 3)

```
In [5]: gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image converted to grayscale!")
print("Grayscale Shape:", gray.shape)
```

Image converted to grayscale!
Grayscale Shape: (512, 512)

```
In [6]: gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image converted to grayscale!")
print("Grayscale Shape:", gray.shape)
```

Image converted to grayscale!
Grayscale Shape: (512, 512)

```
In [7]: plt.figure(figsize=(8, 6))

plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
```

```
plt.show()
```

Original Grayscale Image



```
In [8]: def add_salt_pepper_noise(image, amount=0.05):  
        noisy = image.copy()  
  
        # Salt noise  
        num_salt = int(amount * image.size / 2)  
  
        coords = [  
            np.random.randint(0, i, num_salt)  
            for i in image.shape  
        ]  
  
        noisy[coords[0], coords[1]] = 255  
  
        # Pepper noise  
        num_pepper = int(amount * image.size / 2)  
  
        coords = [  
            np.random.randint(0, i, num_pepper)  
            for i in image.shape  
        ]  
  
        noisy[coords[0], coords[1]] = 0  
  
        return noisy
```

```
noisy_image = add_salt_pepper_noise(gray)

print("Salt-and-pepper noise added successfully!")
```

Salt-and-pepper noise added successfully!

```
In [9]: plt.figure(figsize=(8, 6))

plt.imshow(noisy_image, cmap="gray")
plt.title("Image with Salt-and-Pepper Noise")
plt.axis("off")

plt.show()
```

Image with Salt-and-Pepper Noise



```
In [10]: gaussian = cv2.GaussianBlur(noisy_image, (5, 5), 0)

plt.figure(figsize=(8, 6))
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Filter")
plt.axis("off")
plt.show()
```

Gaussian Filter



```
In [11]: median = cv2.medianBlur(noisy_image, 5)

plt.figure(figsize=(8, 6))
plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")
plt.show()
```

Median Filter



```
In [12]: average = cv2.blur(noisy_image, (5, 5))

plt.figure(figsize=(8, 6))
plt.imshow(average, cmap="gray")
plt.title("Average / Mean Filter")
plt.axis("off")
plt.show()
```

Average / Mean Filter



```
In [13]: laplacian = cv2.Laplacian(gray, cv2.CV_64F)

         laplacian = cv2.convertScaleAbs(laplacian)

         plt.figure(figsize=(8, 6))
         plt.imshow(laplacian, cmap="gray")
         plt.title("Laplacian High-Pass Filter")
         plt.axis("off")
         plt.show()
```

Laplacian High-Pass Filter

